

# Finance Billing Analytics Platform

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## Project Description

Finance Billing Analytics Platform is a web-based financial management system developed using ASP.NET Core 8, Entity Framework Core, SQL Server, MVC, Web API and Clean Architecture principles.

## Objectives

Customer Management, Invoice Management, Payment Management and Dashboard Analytics.

## Technologies Used

ASP.NET Core 8, C#, SQL Server, Entity Framework Core, MVC, Web API, AutoMapper, FluentValidation, JWT Authentication, Swagger and Bootstrap.

## Architecture

User -> MVC Web Application -> Web API -> Application Layer -> Infrastructure Layer -> SQL Server Database

## Domain Layer Folder contains

Contains Entities, Enums, Exceptions and Business Rules.

## Application Layer folder contains

Contains DTOs, Interfaces, Services, Validators and AutoMapper Profiles.

## Infrastructure Layer contains

Contains DbContext, Repositories, Configurations, Identity and Security.

## API Layer contains

Contains Controllers, Middleware, Swagger Configuration and Authentication.

## MVC Layer contains

Contains Controllers, Views, Models and User Interface components.

## Database Relationships

Customer (1:N) Invoice, Invoice (1:N) Payment.

## Project Flow

User -> View -> Controller -> API -> Service -> Repository -> DbContext -> SQL Server -> Response

## Advantages of Clean Architecture

Separation of Concerns, Maintainability, Scalability, Reusability and Testability.

## Complete Folder Structure

FinanceBillingAnalytics

|- FinanceBilling.Domain

|- FinanceBilling.Application

|- FinanceBilling.Infrastructure

|- FinanceBilling.API

|- FinanceBilling.Web